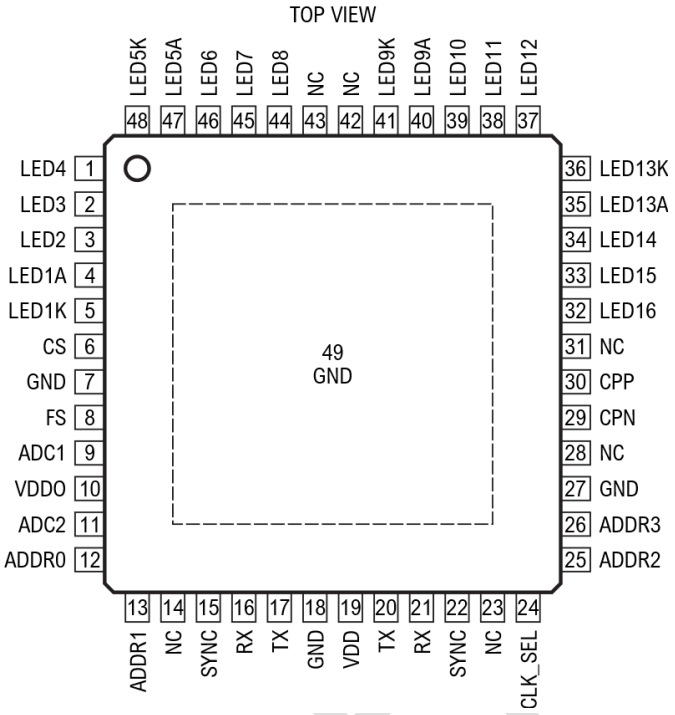
PIN FUNCTIONS



48-LEAD (7mm × 7mm) E-LQFP
EXPOSED PAD (PIN 49) IS GND AND MUST BE SOLDERED TO PCB

VDD (Pin 19): Supply Input. This pin requires local capacitor bypassing to GND. Decouple with one 100nF and one 10nF ceramic capacitor in parallel.

VDDO (Pin 10): VDD Output. Internally connected to the VDD pin for supplying devices that are connected to the general purpose ADC1, ADC2, FS and ADDRx pins. Do not supply the ID803 through this pin. Do not attach any active devices to this pin.

CPP (Pin 30): Charge Pump Positive Terminal. Connect to the top of a 100nF capacitor.

CPN (Pin 29): Charge Pump Negative Terminal, the pin that terminates the 100nF capacitor tied to CPP. Connect an additional 100nF capacitor to GND for local bypassing.

LED1K, LED1A, LED2, LED3, LED4 (Pins 5,4,3,2,1): Bank 1 of 4 Series Switches for Monitoring and PWM control of an LED string. Connect the cathode of the LED string to LED1K. Connect LED1A to the anode of the first LED or series of LEDs to be managed. Connect LED2 to the anode of the second LED or series of LEDs to be managed. Connect LED3 to the anode of the third LED or series of LEDs to be managed. Connect LED4 to the anode of the fourth LED or series of LEDs to be managed. Connect the supply driving the LED string to LED4. Connect the return shunting the LED string current to LED1K. Short adjacent LED pins if not used.

LED5K, LED5A, LED6, LED7, LED8 (Pins 48,47,46,45,44): Bank 2 of 4 Series Switches for Monitoring and PWM control of an LED string. Connect the cathode of the LED string to LED5K. Connect LED5A to the anode of the first LED or series of LEDs to be managed. Connect LED6 to the anode of the second LED or series of LEDs to be managed. Connect LED7 to the anode of the third LED or series of LEDs to be managed. Connect LED8 to the anode of the fourth LED or series of LEDs to be managed. Connect the supply driving the LED string to LED8. Connect the return shunting the LED string current to LED5K. Short adjacent LED pins if not used.

LED9K, LED9A, LED10, LED11, LED12 (Pins 41,40,39,38,37): Bank 3 of 4 Series Switches for Monitoring and PWM control of an LED string. Connect

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the cathode of the LED string to LED9K. Connect LED9A to the anode of the first LED or series of LEDs to be managed. Connect LED10 to the anode of the second LED or series of LEDs to be managed. Connect LED11 to the anode of the third LED or series of LEDs to be managed. Connect LED12 to the anode of the fourth LED or series of LEDs to be managed. Connect the supply driving the LED string to LED12. Connect the return shunting the LED string current to LED9K. Short adjacent LED pins if not used.

LED13K, LED13A, LED14, LED15, LED16 (Pins 36,35,34,33,32): Bank 4 of 4 Series Switches for Monitoring and PWM control of an LED string. Connect the cathode of the LED string to LED13K. Connect LED13A to the anode of the first LED or series of LEDs to be managed. Connect LED14 to the anode of the second LED or series of LEDs to be managed. Connect LED15 to the anode of the third LED or series of LEDs to be managed. Connect LED16 to the anode of the fourth LED or series of LEDs to be managed. Connect the supply driving the LED string to LED16. Connect the return shunting the LED string current to LED13K. Short adjacent LED pins if not used.

CS (Pin 6): Current Sense Input. Connect to LED1K with a sense resistor to GND for current measurement. Float this pin when unused.

FS (Pin 8): Fail-Safe Configuration Pin. The ID803 enters fail-safe when the configurable communication watchdog timer expires. The state of this pin defines the state of the switches during fail-safe operating mode. A logic low state sets all switches to be open (OFF) in fail-safe mode. A logic high state sets all switches to be closed (ON) in fail-safe mode. An internal typical 200kΩ pull-down resistor sets a logic low state when this pin is floating.

SYNC (Pins 15,22): PWM Period Synchronization Pin. This pin is configurable to be the synchronization source or the receiver of synchronization which is configurable through OUTCTRL(0x01) register. Connect to the SYNC pin on other ID803 devices for PWM synchronization. An internal typical 500kΩ pull-down resistor sets a logic low state when this pin is floating. The two SYNC pins are internally connected to allow for ease of PCB layout in connecting additional ID803 devices.

RX (Pins 16,21): UART Receive Data Input. Connect to RX of the CAN transceiver or UART and to the RX pin of additional ID803 devices. The two RX pins are internally connected to allow for ease of PCB layout in connecting additional ID803 devices.

TX (Pins 17,20): UART Transmit Data Output. Connect to TX of the CAN transceiver or UART and to the TX pin of additional ID803 devices. The two TX pins are internally connected to allow for ease of PCB layout in connecting additional ID803 devices.

ADDR0, ADDR1, ADDR2, ADDR3 (Pins 12,13,25,26): Device Address Configuration Pins. ADDR3 is the MSB and ADDR0 is the LSB. Tie to GND for logic low. An internal typical 200kΩ pull-up resistor sets a logic high state when this pin is floating.

ADC1, ADC2 (Pins 9,11): General Purpose ADC Input Pins. If unused, tie to GND through no more than 100k and no less than 10k of resistance.

CLK_SEL (Pin 24): System Clock Selection Pin. A logic low state selects 8MHz system clock operation (500kHz UART baud rate). A logic high state selects 16MHz system clock operation (1MHz UART baud rate). An internal typical 200kΩ pull-up resistor sets a logic high state when this pin is floating.

GND (Pins 7,18,27, Exposed Pad 49): Supply Ground. Tie this pin to ground plane on the top-layer of the PCB. The local bypass capacitor from VDD should be terminated to this plane.

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TYPICAL CHARACTERISTICS

Figure 1 VDD Operating current vs Temperature	Figure 2 VDD UVLO vs Temperature
To be updated Figure 3 VDD OVP vs Temperature	Figure 4 LEDx Overvoltage Threshold vs Temperature
Figure 5 Switch ON-Resistance vs Temperature	Figure 6 Oscillator frequency vs Temperature

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FAULTS AND DIAGNOSTICS

Error! Reference source not found. summarizes the ID803 behavior under each fault detection and protection.

Table 4. Faults and Diagnostics Description

CONDITION	DETECTION	FAULT Reaction	STATUS BIT	CONFIGURABLE FAULT RESPONSE
VDD is below UVLO	VDD < 3.92V(typ)	Yes	PWR	N/A
	Undervoltage condition of system supply voltage is monitored at VDD pin. If the VDD voltage falls below a predefined threshold, 3.92V typ, the device transitions to Unpowered state, and all switches are set to open.			
VDD Overvoltage	VDD > 5.77V typ	No	VDD_OV_FLT	EN_VDD_OV
	If VDD voltage exceeds a predefined threshold, 5.77V typ, the switch between CPN and VDD is opened if the highest LEDx voltage is less than 4.2V and the fault bit, VDD_OV_FLT, is set to “1”.			
Resistive FET Detection	VLEDx > VTH_SHORT	No	FLT_RESFET[16:1]	NA
	If the differential voltage of each internal switch is greater than the LED Short Threshold ($V_{TH\text{-}SHORT}$) right before the end of a switch ON pulse, then a fault is reported by setting the corresponding FLT_RESFET bits to “1”.			
LED Open Detection	VLEDx > VTH_OPEN	Yes	FLT_OPEN_OR_DRV[16:1]	NA
	ID803 monitors open condition across each SW FET. If the voltage across each switch exceeds LED Open Threshold ($V_{TH\text{-}OPEN}$), the switch FET is closed and all the corresponding bits in the FLT_OPEN_OR_DRV are set to “1”.			
LED Short Detection	VLEDx < VTH_SHORT	No	FLT_SHORT[16:1]	NA
	ID803 monitors short condition across each SW FET and reports if the voltage across each switch decreases below the LED Short Threshold ($V_{TH\text{-}SHORT}$) at the end of LED ON, a corresponding register bit in FLT_SHORT register is set to “1”.			
Driver health diagnostic	NA	No	FLT_OPEN_OR_DRV[16:1]	DRV_CHK
	To check the health of the SW FET driver, the DRV_CHK bit in the OUTCTRL register can be set by MCU. Upon setting this bit, all the gate driver outputs are disconnected from the output switches and the gate biases are pulled low through internal pull-down resistor. If all the switches are correctly turned off and are at high-z, all the corresponding bits in the FLT_OPEN_OR_DRV are set to "1".			
Matrix SW POR	NA	Yes	FLT_MATRIX_POR[16:1]	NA
	If the floating supply in the switch module falls below a defined threshold, the switch is set to open. This mechanism is triggered if the internal supply to the regulator or the VDD supply, or the charge pump fails. ID803 sets the corresponding switch open and all the corresponding bits in the FLT_MATRIX_POR register are set to "1".			
LED Current Monitoring	NA	No	CS	CSEN, CSGAIN
	If the CSEN bit is set to “1” in the SYSCFG register, the voltage across CS pin and GND is measured using an ADC. The measured results are stored in the CS register.			
UART Communication Watchdog	Not Receiving time valid UART > CMWTAP	Yes	OPMODE	CMWEN, CMWTAP
	If CMWEN bit in SYSCFG register is set to “1” and a valid UART transaction for more than a pre-programmed time in CMWTAP bits is not received, the device enters the FAIL-SAFE state, and an associated error bit is set in the status register. In FAIL_SAFE state, all switches are OFF if FS pin is low, all switches are ON if FS pin is high.			
Internal UART Watchdog Check	Not Receiving time valid UART > CMWTAP/8 right after start-up	No	OPMODE	CMWEN, CMWTAP
	A health check of the UART watchdog can be performed upon power up. MCU should wait at least 125ms/250ms (dependent on CLK_SEL pin) without any UART communication after startup. If the UART watchdog is not functional, ID803 will stay at DIAG1 mode, otherwise ID803 will enter DIAG2 mode. The switches are kept OFF in DIAG1 mode			
PWM Monitoring	NA	No	PWM_ERR, PWM_MISCOUNT	NA
	The diagnostic PWM counter checks the output of each channel PWM counter in a round robin to make sure the output of each PWM counter is programmed. If a PWM error count occurs, PWM_ERR bit is set to “1” in the STATUS register and a bit for a corresponding PWM counter is set to “1” in the PWM_MISCOUNT registers.			
FS pin status	NA	No	FS_PIN	NA
	During FAIL-SAFE state, switches are set according to the FS-pin voltage, all switches shall be set to open if the voltage on the FS-pin is lower than $V_{IL\text{-}TH}$, and shall be set to close if the voltage on the FS-pin is more than $V_{IH\text{-}TH}$. The FS-pin			

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Table 4. Faults and Diagnostics Description

CONDITION	DETECTION	FAULT Reaction	STATUS BIT	CONFIGURABLE FAULT RESPONSE
	can be connected to GND or the VDD. There is an FS_PIN bit in the STATUS1 register that shows the voltage level of the FS pin.			
Charge pump Voltage Monitoring	VCP < V_{CPTH-F}	Yes	CHPMP_ERR	NA
	The charge pump voltage is monitored. If the charge pump voltage is decreased below a threshold with t_{CP_GF} deglitch time, all the switches are set to open position. The CHPMP_ERR bit is set to “1” in the STATUS2 register if a charge pump error occurs.			
Internal Supply Monitoring	Internal supply < UVLO threshold	Yes	PWR	NA
	When 1.8V internal bias reduces below a specified threshold, it generates a POR that resets digital and sets the output switches to open. The PWR bit is then set to default value of “1”.			
Bandgap ADC	BGR > 0xA4 or BGR < 0x90	No	BGR	NA
	The ID803 has secondary bandgap (typical 1.2V) and digitizes the secondary bandgap voltage with the 2V reference generated by the primary bandgap. The ADC result is stored in BGR register.			
Thermal Limit	T_J > 175°C typically	Yes	TSD	TSDEN
	If the measured DIETEMP by ADC is higher than 175°C, the TSD bit in STATUS2 register is set. If thermal shutdown is detected, switches are kept turned OFF when TSDEN is configured to 11b (if there is no OPEN detected) until DIETEMP is below 160°C and the bit should be cleared by MCU			

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